An Enumeration of Results Concerning Approximation of Nonstandard Varieties

Alexander Michos

October, 2023

1 Two Methods in Real-Valued Differentiation & Inversion

To isolate terms from the Taylor expansion of a function f, it is standard to find path integrals about the origin in the complex plane;

$$f^{(n)}(0) = \frac{n!}{2\pi}\int_{-\pi}^{\pi} f\left(e^{i\theta}\right) e^{-ni\theta} d\theta$$

However this formulation, and the functional inversion soon to come, are thus bound to the complex numbers. While true, analytic functions are nicely described, under a wealth of results, by contours about their roots and poles, is the same true for the real line?

Calling in the Legendre Polynomials $L(n,x)$, we obtain the immediate expansion;

$$\hat{f}_h^N(x) = f(0) + \sum_{m=1}^{N} \frac{x^m}{m!}\frac{m!}{h^m}\frac{\int_{-1}^{1} L(m,t) f(ht)\, dt}{\int_{-1}^{1} L(m,t)\, t^m dt}$$

Which is nothing more than;

$$\frac{1}{h^n}\int_{-1}^{1} L(n,x)\left(f(0) + \frac{f'(0)}{1!}xh + \dots \frac{f^{(n+1)}(0)}{(n+1)!}(xh)^{(n+1)}\right) dx \to \frac{f^{(n)}(0)}{n!}\int_{-1}^{1} L(n,x)\, x^n + h\int_{-1}^{1} L(n,x)\, R_n(x)\, dx$$

For sufficiently small h all higher order terms vanish, and all lower order terms are integrated to 0 by Legendre Polynomial orthogonality. Standard identities bring us to;

$$\hat{f}_h^N = f(0) + \frac{1}{2}\sum_{m=1}^{N}\left(\frac{x}{2h}\right)^m (2m+1)\,\mathrm{nCr}(2m,m)\int_{-1}^{1} L(m,t) f(ht)\, dt$$

The complexities arise due to the instability of this expansion, the smaller the h, the worse the interval of convergence but the better the accuracy near the origin, and the larger the N the worse the numerical stability.

After all, $\sum_{m=1}^{N} \mathrm{nCr}(2m,m)\left(\frac{x}{2h}\right)^m$ converges only for $\left|\frac{x}{2h}\right| \leq \frac{1}{4}$, which is less than the $|x| < 1$ interval of the requisite integral. Though for the interested reader we can construct an ‘inversion kernel’ via an integral formula for the central binomial coefficients.

$$\sum_{n=1}^{\infty} L(n,t)\,\mathrm{nCr}(2n,n)\, x^n = \frac{1}{\pi}\int_{\frac{1}{4}}^{\infty} \frac{1}{\sqrt{4u-1}\sqrt{u^2 - 2xtu + x^2}} du - 1$$

But, if we cast away this unhelpful divergence, there are still more applications to explore. The famous Lagrange Inversion theorem gives an explicit construction of the inverse function to f in terms of complex residues. However, in the spirit of isolating this behavior to real integrals we may take;

$$a + \sum_{n=1}^{\infty} D^{(n-1)}\left[\left(\frac{x-a}{f(x)-f(a)}\right)^n\right]\frac{(z-f(a))^n}{n!}$$

$$f^{-1}(p) = a + \frac{h}{2}\sum_{n=1}^{N}\int_{-1}^{1} L\left(n-1,t\right)\mathrm{nCr}\left(2n,n\right)\left(\frac{t}{f\left(a+ht\right)-f\left(a\right)}\frac{p-f\left(a\right)}{2}\right)^{n}dt$$

The combination of Legendre Polynomials and the Lagrange Inversion theorem has proved surprisingly effective up to a truncation and moderately loose h. Establishing the optimal number of terms to take or the granularity of the scale factor remain unresolved inquiries.

We now pursue a related yet distinct approach to real-valued inversion. Given any orthogonal function set or limiting distribution to construct the delta function, the most trivial approach to real-valued inversion would be simply; (where I contains the requisite inverse)

$$f^{-1}\left(p\right) = \frac{\int_I \delta\left(f\left(t\right)-p\right)tdt}{\int_I \delta\left(f\left(t\right)-p\right)dt}$$

Being mindful of the interplay between the chain rule properties and inverses: $\delta\left(f\left(t\right)-z\right) = \frac{\delta\left(t-f^{-1}(z)\right)}{f'\left(f^{-1}(z)\right)} = \delta\left(t-f^{-1}\left(z\right)\right)f^{-1\prime}\left(z\right)$. Such a quotient of integrals is taken to cancel the 'chain rule factors' of the delta function and also conveniently eliminates the need for a normalization factor for the delta convergents (δ_ϵ).

A nice choice for such a delta function convergent is the following limit,

$$\lim_{\epsilon\to 0}\frac{\int_a^b \frac{c^k}{\left((f(t)-z)^2+\epsilon^2\right)^N}tdt}{\int_a^b \frac{c^k}{\left((f(t)-z)^2+\epsilon^2\right)^N}dt} = f^{-1}\left(z\right)\ ;\ \forall N\geq 1,k$$

Should a fraction of integrals be undesirable, we can create the appropriate chain-rule-factor in the numerator as in the following simpler double-integral;

$$\lim_{\epsilon\to 0}\left(\arctan\left(\frac{b}{\epsilon}\right)-\arctan\left(\frac{a}{\epsilon}\right)\right)^{-2}\int_a^b \frac{\epsilon}{\left(f\left(t\right)-z\right)^2+\epsilon^2}tdt\int_a^b \frac{\epsilon}{\left(f\left(t\right)-z\right)^2+\epsilon^2}f'\left(t\right)^2dt$$

The arctan-based delta function is used here again, but any limiting distribution or series definition will work equivalently.

A third possibility, relying on the derivative-of-inverses identity, is the following;

$$\frac{f^{-1}\left(q\right)^2-f^{-1}\left(p\right)^2}{2} = \int_p^q f^{-1}\left(z_0\right)f^{-1\prime}\left(z_0\right)dz_0 = \lim_{\epsilon\to 0}\int_p^q\int_a^b \delta_\epsilon\left(f\left(t\right)-z_0\right)tdtdz_0$$

Which is perhaps the simplest of any of the expressions introduced here, but which regrettably encloses the inverse function in a square.

Constructions such as these are likely computationally impractical given the limits of numerical integration. However, given the generality with which one can define the delta function, and ensuring to normalize the expression for the appropriate interval where an inverse lies, perhaps there are theoretical applications.

I simply cannot help but plug in common inverse problems and find limits for logarithms, arc-trig, Lambert-W functions, and so forth. In this sense, real-valued methods for differentiation and inversion may be a fruitful study.

2 Several Refinements on the Isoperimetric Inequality

The *isoperimetric inequality*, confined to the R^2 plane, is a classic result which formalizes the notion that the circle possesses the least perimeter for a given surface area. All other closed, continuous curves must have an arc length L, and area A which exceeds this benchmark;

$$L^2 \geq 4\pi A$$

With this in hand, there is nothing to stop one from taking integrals of the form $\frac{1}{2\pi}\int_0^{2\pi}\Phi(t)e^{kit}dt$, extracting the coefficients needed to compute either the area or arc-length and calling the approximation complete there. However, this computationally intelligent approach lacks the elegance in the relationships between length, area, and radius.

Suppose that we have described our curve in polar form $R(\theta)$ that resembles a circle of radius r up to perturbation of magnitude α and β by sin and cos terms. How can we improve the 'isoperimetric approximation' to L to be more accurate?

$$L^2 = \left(\int_0^{2\pi} \sqrt{r^2+\alpha^2+\beta^2+2r\alpha\sin\theta+2r\beta\cos\theta d\theta}\right)^2$$

Gently Taylor expanding;

$$\left(\int_0^{2\pi} \sqrt{r^2+\alpha^2+\beta^2} + \frac{1}{2\sqrt{r^2+\alpha^2+\beta^2}}\left(2r\alpha\sin\theta+2r\beta\cos\theta\right) - \frac{\left(2r\alpha\sin\theta+2r\beta\cos\theta\right)^2}{8\left(r^2+\alpha^2+\beta^2\right)^{\frac{3}{2}}} d\theta\right)^2 \sim L^2$$

$$\left(\sqrt{c^2+\alpha^2+\beta^2}2\pi - \frac{\pi 4c^2\alpha^2+\pi 4c^2\beta^2}{8\left(c^2+\alpha^2+\beta^2\right)^{\frac{3}{2}}}\right)^2$$

Substituting in Variance and Area;

$$V = \frac{1}{2\pi}\int_0^{2\pi} R(\theta)^2\,d\theta - \left(\frac{1}{2\pi}\int_0^{2\pi} R(\theta)\,d\theta\right)^2 = \frac{\alpha^2+\beta^2}{2}$$

$$A = \frac{1}{2}\int_0^{2\pi} R^2 d\theta = \frac{1}{2}\left(2\pi c^2+\pi\alpha^2+\pi\beta^2\right)$$

We conclude this basic arclength approximation with;

$$L^2 \approx 4\pi A \approx \pi\left(\frac{\left(2A^2+3\pi^2V^2+3A\pi V\right)^2}{\left(A+\pi V\right)^3}\right)$$

Should the variation in the radius of the curve vanish to 0, we are left with the standard 'isoperimetric approximation', and in cases where a perturbation of period 2π is added to the curve, this formula works fantastically.

In cases where this does not hold, the following approach was employed; considering

$$\mathcal{L}(\epsilon) = \left(\int_0^{2\pi} \sqrt{R(\theta)^2+\epsilon R'(\theta)^2} d\theta\right)^2$$

$$\left(\int_0^{2\pi} R(\theta)\,d\theta\right)\left(\int_0^{2\pi} R(\theta)\,d\theta + \int_0^{2\pi}\frac{R'(\theta)^2}{R(\theta)}d\theta - \frac{1}{4}\int_0^{2\pi}\frac{R'(\theta)^4}{R(\theta)^3}d\theta\right) + \frac{1}{4}\left(\int_0^{2\pi}\frac{R'(\theta)^2}{R(\theta)}d\theta\right)^2$$

Although more applicable to a wider variety of polar curves, it is not readily expressible in terms of quantities such as 'radius variance' or area.

3 Polynomial Refinements to a Cauchy-Schwarz Approximation

It is an exceedingly fun exercise to use polynomial approximations to correct for 'deficits' in common inequalities. Take for instance Cauchy-Schwarz, which informs us that;

$$\left(\int_0^1 f(x)^2\,dx\right) - \left(\int_0^1 f(x)\,dx\right)^2 \geq 0$$

But this quantity is really not very close to 0, so we describe this 'variance' measure of sorts for functions on [0,1] as an operator on polynomials. Thus, we take f to be polynomial and find the difference is approximated by;

$$\approx \frac{1}{60}\int_0^1 f'(x)^2\,dx + \frac{1}{15}\left(\int_0^1 f'(x)\,dx\right)^2$$

Or the significantly more unhelpful but more accurate;

$$\approx \left(\frac{1}{12}\int_0^1 f'(x)^2\,dx - \frac{1}{360}\int_0^1 f'(x)\,f'''(x)\,dx\right) + \left(-\frac{1}{1008}\int_0^1 f''(x)^2\,dx - \frac{23}{5040}\left(\int_0^1 f''(x)\,dx\right)^2\right)$$

The statement 'equality holds when f is constant' now takes on a more interesting meaning; that of the 'higher order terms' vanishing as a result of the lack of variation in f. Of course, any function that is approximated by polynomials will be vastly better approximated by the above two formulations than by the 'order-0' Cauchy-Schwarz approximation to the difference of integrals.

3.1 Additional Inequality

In an attempt to turn the approximations of above into true inequalities we may examine this nice application of the Cauchy–Schwarz inequality;

$$\left(\frac{1}{R}\int_0^R f(x)\,dx\right)^2 = \left(\sum_n \frac{a_n^n R^n}{n+1}\right)^2 \le \left(\sum_n a_n^2 R^{2n}\right)\left(\sum_n \frac{1}{(n+1)^2}\right) \le \frac{1}{2\pi i}\int_{|z|=R} \frac{1}{z}\,|f(z)|^2\,dz \cdot \frac{\pi^2}{6}$$

Which gives a much more rigorous connection between the squared integral of f and the integral of f squared, though here trapped in a contour integral.

4 Differentiating by Integrating

The *Lanczos Derivative*, defined below, is an example of the interesting process of Differentiation by Integration;

$$f'(x) = \lim_{h\to 0}\frac{3}{2h^3}\int_{-h}^{h} t f(x+t)\,dt = 3\int_0^1 t\lim_{h\to 0}\frac{f(x+th) - f(x-th)}{2h}dt$$

As is seen in the second rearrangement, the process of differentiation by integration harnesses an infinitude of individual derivative estimates; using the precision in integration to augment a base central difference approximation, for example.

However, assuming perfect integration, we can far exceed the accuracy of standard Lanczos by picking (not quite Legendre) polynomials such that;

$$\int_0^1 p_k(x)\,x^j\,dx = \left\{\begin{array}{ll} 0 & j \in [2, k+1] \\ \neq 0 & j = 1 \end{array}\right\}$$

For a fixed h, as many of the higher order terms in the Taylor expansion of f will vanish under integration, helping to greatly accelerate convergence.

$$f'(0)_h = \frac{1}{\int_0^1 p_k(x)\,x dx}\int_0^1 p_k(x)\,\frac{f(xh) - f(0)}{h}dx$$

A sampling of these polynomials are given;

$$p_2(x) = x^2 - \frac{4}{3}x + \frac{2}{5}$$

$$p_3(x) = x^3 - \frac{15}{8}x^2 + \frac{15}{14}x - \frac{5}{28}$$

Leading to formulas such as;

$$f'(0) = \lim_{h\to 0} \frac{180}{h^2} \left(\int_0^h \left(\frac{t^2}{h^2} - \frac{4}{3h}t + \frac{2}{5} \right) f\left(t\right) dt - f\left(0\right) \frac{h}{15} \right)$$

Or, the outrageous yet astoundingly accurate $O(h^7)$;

$$p_6(x) = \frac{4}{429} - \frac{24}{143}x + \frac{150}{143}x^2 - \frac{40}{13}x^3 + \frac{60}{13}x^4 - \frac{24}{7}x^5 + x^6$$

$$f'(x)_h = 168168 \int_0^1 \left(\frac{4}{429} - \frac{24}{143}t + \frac{150}{143}t^2 - \frac{40}{13}t^3 + \frac{60}{13}t^4 - \frac{24}{7}t^5 + t^6 \right) \frac{f\left(x+th\right) - f\left(x\right)}{h} dt$$

Note that in each integrand we are *not* integrating the quantity $\frac{f(x)-f(0)}{x}$, which would defeat the purpose of this exercise; avoiding a divided difference.

Noting the inherent benefits in a symmetric integral, we can of course generalize the Lanczos Formula from above to arbitrary degree;

$$f'\left(x\right) = -\frac{105}{8h} \int_{-1}^{1} \left(t^3 - \frac{5}{7}t \right) f\left(ht + x\right) dt + \sim f'''''(x)h^4$$

For a process corrupted by Brownian error, for instance, simply finding the integral under a path can be an effective means of computing its derivative over a region. Though, it ought to be noted that methods of differentiating noisy signal data, such as through Total Variation De-Noising, are all well established for this purpose.

For higher derivatives, we can adjust the defining properties of p_k to capture say the 2nd term in the Taylor Series, however in *Differentiation by integration using orthogonal polynomials, a survey*, the authors display simple, elegant, and effective means of higher-order differentiation by integration using standard orthogonal polynomials.

5 Continuous Single-Variable Total Variation Denoising

A fascinating study in numerical analysis and signal processing is Total Variation Denoising; a technique to balance the fidelity, or accuracy, of a process applied to noisy data against its variation, so as to constrain the output and produce a useable result. For example, taking the raw finite difference for a noisy signal may be the most *accurate* derivative, but it will contain so much oscillatory noise so as to be unhelpful, and hence must be balanced against a low-variation estimate; where 'balance' is achieved with a tunable weighting factor a.

Let us exercise these techniques to differentiate a single-variable, differentiable function on the interval [0,N]. What we obtain is an alternative 'definition' of the derivative, which can capture an adjustable amount of variation in the target function. We seek to minimize;

$$\int_0^N y'^2 + a\left(f'\left(t\right) - y\right)^2 dt$$

Solving the standard Euler-Lagrange gives;

$$Y\left(t, c_1, c_2\right) = \frac{1}{2}\sqrt{a}e^{-\sqrt{a}t} \int_0^t e^{s\sqrt{a}} f'\left(s\right) ds - \frac{\sqrt{a}}{2} e^{\sqrt{a}t} \int_0^t e^{-s\sqrt{a}} f'\left(s\right) ds + c_1 e^{\sqrt{a}t} + c_2 e^{-\sqrt{a}t}$$

$$-\sqrt{a} \int_0^t \sinh\left(\left(t-s\right)\sqrt{a}\right) f'\left(s\right) ds + c_1 e^{\sqrt{a}t} + c_2 e^{-\sqrt{a}t}$$

Call the integral component $Y_0(t)$ for convenience. Now, we seek to minimize the original functional with respect to $c1, c2$; completing the differentiation and algebra;

$$c_1 = c_2 = \frac{-\left(-\int_0^N 2a\left(f'\left(x\right) - Y_0\left(x\right)\right) e^{-\sqrt{a}x} dx - \sqrt{a} 2 \int_0^N \left(\frac{d}{dx} Y_0\left(x\right)\right) e^{-\sqrt{a}x} dx\right)}{\int_0^N 4a\left(e^{-\sqrt{a}x}\right) e^{-\sqrt{a}x} dx}$$

This quotient reduces to;

$$\frac{-\sqrt{a}}{2}\frac{\int_0^N f'(s)\left(e^{-\sqrt{a}s}+e^{-\sqrt{a}2N}e^{s\sqrt{a}}\right)ds}{e^{-2\sqrt{a}N}-1}$$

Giving the 'tunable derivative' (interpolating between average growth and differentiation) under the ideas of TV as follows.

$$\boxed{f'^{[a]}=-\sqrt{a}\int_0^t \sinh\left((t-s)\sqrt{a}\right)f'(s)\,ds-\sqrt{a}\frac{\int_0^N f'(s)\left(e^{-\sqrt{a}s}+e^{-\sqrt{a}2N}e^{s\sqrt{a}}\right)ds}{e^{-2\sqrt{a}N}-1}\cosh\left(\sqrt{a}t\right)}$$

By taking a to 0, we stipulate that no variation can be present and hence $f'^{[0]}$ takes on the average rate of change;

$$\lim_{a\to 0} f'^{[a]}=-2\int_0^N f'(s)\,ds\lim_{a\to 0}\frac{\sqrt{a}}{e^{-2\sqrt{a}N}-1}=\frac{f(N)-f(0)}{N}$$

Or, as we weight the accuracy term more and more, heavily, $f'^{[\infty]}$ will approach the exact derivative;

$$\mathcal{L}\left\{f'(x)\right\}=\mathcal{L}\left\{-a\int_0^t \sinh\left((t-s)a\right)f'(s)\,ds-a\frac{\int_0^N f'(s)\left(e^{-as}e^{2aN}+e^{sa}\right)ds}{1-e^{2aN}}\cosh(at)\right\}$$

$$=\lim_{a\to\infty}-\frac{a^2}{s^2-a^2}\mathcal{L}\left\{f'(t)\right\}+a\int_0^N f'(s)\,e^{-as}ds\frac{s}{s^2-a^2}$$

$$\mathcal{L}\left\{f'(x)\right\}=\mathcal{L}\left\{f'(x)\right\}$$

6 Intermediary Taylor Series

Suppose, once again we aspire to construct $f(x)$ from a finite set of basis functions, $\phi_k(x)$. Our (quite trivial) observation is that by expressing f and $n-1$ ϕ_k's to their nth degree Taylor Expansion, standard Gaussian Reduction can be employed to express f as an approximate linear combination in ϕ.

$$f(x)\sim\sum_k c_k\phi_k(x)\ \ ;\ \ \left[\vec{\phi}_{1TS},\vec{\phi}_{2TS}...\vec{\phi}_{nTS}\right]\vec{c}=\vec{f}_{TS}$$

Where these TS vectors are column vectors containing the coefficients.

This computation is highly inspired by the process of numerical differentiation, obtaining constants c_k, such that as much of the TS expansions cancel.

$$\sum_k c_k f_{TS}(x-kh)=f'(x)$$

And, the coefficients of Central Difference and 5-Point Stencil are indeed constructed in exactly this fashion; *match the TS on both sides to achieve accuracy of approximation.* Take any set of functions, $\sin(x),\sin(2x),x\cos(x),\ln(x),\sinh(7x+3)$ etc, any target function $1,x,e^x$ etc, and after inverting the matrix containing the Taylor coefficients of the former, one obtains a linear (approximate) construction of the latter.

Let's indulge some examples where Taylor Series are used as an intermediary to produce functional approximations; (For $x\sim 0$)

$$\frac{-1.5+3e^x-0.5e^{2x}-e^{-x}}{3}\sim x$$

$$\frac{\frac{11}{3}\sin(x)+\frac{2}{3}\sin(2x)-4\cos(x)\,x}{x}\sim\cosh(x)$$

$$\frac{8}{5}\cosh(x)-\frac{4}{5}\cosh(2x)+\frac{8}{35}\cosh(3x)-\frac{1}{35}\cosh(4x)\sim 1$$

$$-\frac{61}{45}\cosh(x)+\frac{169}{90}\cosh(2x)-\frac{3}{5}\cosh(3x)+\frac{7}{90}\cosh(4x)\sim x^2$$
$$\frac{106}{45}\cosh(x)-\frac{169}{90}\cosh(2x)+\frac{6}{10}\cosh(3x)-\frac{7}{90}\cosh(4x)\sim\cos(x)$$

Sines, cosines, and their hyperbolics are often effective as half of their coefficients are 0, meaning to ensure a set of n linearly independent Taylor Series vectors, we infact pursue up to the $2n$ degree Taylor approximation. Also;

$$\frac{-\ln\left(\frac{38086}{10^4-1}\right)+18\ln(10)}{\sqrt{163}}\sim\pi+8\cdot10^{-10}$$

7 Discrete Orthogonal Series

In the quest for an Orthogonal expansion of a function in terms of basis functions, we inevitably define an inner product and conduct Gram-Schmidt. Here, define a basis of $\{2^{-n},3^{-n},4^{-n}...\}$ and a *discrete* inner product;

$$[f,g]=\sum_{n=1}^{\infty}f(n)\,g(n)$$

It is highly likely, seeing the trivial nature of this construction, that these definitions are well established. In any case, it is intriguing to parallel the ubiquitous polynomial basis on an *integral* inner product, with these discrete geometric bases (DGB). The first few DGB's are;

$$f_1(x)=2^{-x}$$

$$f_2(x)=3^{-x}-\frac{3}{5}2^{-x}$$

$$f_3(x)=4^{-x}-\frac{80}{77}3^{-x}+\frac{15}{77}2^{-x}$$

$$f_4(x)=5^{-x}-\frac{55}{38}4^{-x}+\frac{220}{399}3^{-x}-\frac{5}{114}2^{-x}$$

What is fascinating, and once again trivial, is that;

$$\sum_{n\geq1}q^{-n}f_{\geq q}(n)=0$$

And that, for the generating function H of h;

$$\left[h(x),q^{-x}\right]=H\left(\frac{1}{q}\right)$$

The typical orthogonal construction of;

$$h\sim\sum\frac{[h,f_k]}{[f_k,f_k]}f_k$$

Allows for the conveient approximation of;

$$\sum_{n\geq1}h(n)\,g(n)$$

Where the decomposition of h is known, or even more conviently if the decomposition of h and g is known. Once decomposed into basis components we of course have $[h,g]=h^TFg$, where F contains the inner product of each DGB, trivially calculated by a geometric series.

Peculiar functions, such as $1/x!$, have convient intrepretations under the summatory inner product; more so than in the continous case.

$$\frac{1}{x!} \approx f_1(x)\frac{e^{\frac{1}{2}}-1}{\frac{1}{3}} + f_2(x)\frac{\left(e^{\frac{1}{3}}-1\right)-\frac{3}{5}\left(e^{\frac{1}{2}}-1\right)}{\frac{1}{200}} + f_3(x)\frac{\left(e^{\frac{1}{4}}-1\right)-\frac{80}{77}\left(e^{\frac{1}{3}}-1\right)+\frac{15}{77}\left(e^{\frac{1}{2}}-1\right)}{\frac{4}{88935}} +$$

$$+f_4(x)\frac{\left(e^{\frac{1}{5}}-1\right)-\frac{55}{38}\left(e^{\frac{1}{4}}-1\right)+\frac{220}{399}\left(e^{\frac{1}{3}}-1\right)-\frac{5}{114}\left(e^{\frac{1}{2}}-1\right)}{\frac{1}{3820824}}$$

8 Mittag-Leffler Expansion

Suppose that for meromorphic f with simple poles, at a_n, with residue b_n we have;

$$f(z) = f(0) + \sum_{n=1}^{\infty} b_n \left(\frac{1}{z-a_n} + \frac{1}{a_n}\right)$$

At $z = 0$, we have $f(0) = f(0)$ trivially, whilst at every pole we have matching residues on both sides. Construction of this Mittag-Leffler series requires that f be bounded on paths taken between any of the poles, however;

Observation: If f is unbounded, the Mittag-Leffler series acts as a means of approximation for f in the region containing the set of poles. Thus, the stage is set to explore a family of approximations centered about the poles to a target function.

For $\Re z \leq 0$, for example we have the peculiar;

$$\frac{1}{\zeta(z)} \sim -2 + \frac{1}{\pi}\sum_{n=1}^{\infty}\left(\frac{(-1)^n}{(2n)!}\frac{(2\pi)^{(1+2n)}}{\zeta(1+2n)}\right)\left(\frac{1}{z+2n} - \frac{1}{2n}\right)$$

Which can be augmented, as necessary, through inclusion of the non-trivial zeros following;

$$z_n \sim 1/2 + i\frac{\pi(8n-7)}{4W\left(\frac{8n-7}{8e}\right)}$$

For the Lambert-W function. The Gamma function also can be *approximated* along the region with poles ($\Re z < 0$);

$$\Gamma(z) \sim c + \sum_{k=1}^{\infty}\frac{(-1)^k}{k!}\left(\frac{1}{z+k} - \frac{1}{k}\right) + \frac{1}{z}$$

For an appropriately chosen c, which best matches the height of the function where desired. In the region around $x = -1/2$, a value of $c = 1-\phi$ gives a remarkably accurate estimate of Γ. Alternatively, for functions with pole behavior, we can use the MLE to handle the general pole and residue shape, and fit regressions to the more predictable behavior of c;

$$\sum_{k=0}^{\infty}\frac{(-1)^k}{k!}\left(\frac{1}{x+k}\right) + \frac{1}{0.373157}e^{0.373157x-2.78718} + 0.0266$$

Using the idea of applying a regression to;

$$|f(z) - M_{ittag}(z)|$$

To fine-tune the local behavior, we arrive at; ($\Re z < 0$)

$$\frac{1}{\zeta(z)} \sim 0.026088z - 2.00953 + \frac{1}{\pi}\sum_{n=1}^{\infty}\left(\frac{(-1)^n}{(2n)!}\frac{(2\pi)^{(1+2n)}}{\zeta(1+2n)}\right)\left(\frac{1}{z+2n} - \frac{1}{2n}\right)$$

Accurate, despite only 'matching' residues, to around 2-4 decimal places in the appropriate region.

9 Gaussian Approximations & Limiting Behavior

In is interesting to note that;

$$e^{-x^2} \sim \left(\frac{\sin\left(\frac{x}{\sqrt{2}}\right)}{\frac{x}{\sqrt{2}}}\right)^{12} \sim \left(\frac{\sin\left(\frac{x}{2}\right)}{\frac{x}{2}}\right)^{24} \sim \operatorname{sech}\left(\frac{x}{2}\right)^{8}$$

The remarkable degree of accuracy of these approximations warrants a closer elementary examination. For any 'bell-curve look-alike' f which satisfies $f(0) = 1, f'(0) = 0$, it is trivial to note that;

$$f\left(x/a\right)^{-\frac{2a^2}{f''(0)}} \to e^{-x^2}$$

And for $f(h), f''(h) \neq 0$, more generally we have;

$$\lim_{a\to\infty}\left(\frac{f\left(\frac{x}{a}+h\right) - f'\left(h\right)\frac{x}{a}}{f\left(h\right)}\right)^{-\frac{2a^2 f(h)}{f''(h)}} = e^{-x^2}$$

Verifiable by Lohpital and standard limit techniques. Let us enter into an additional interlude to prove the CLT via this convergence of arbitrary distributions to the Gaussian.

9.1 Related CLT Exercise

Suppose f is a distribution on $(-\infty, \infty)$ with mean 0 and variance of 1. Since addition of probability follows Fourier convolution, we claim:

$$\frac{1}{\sqrt{n}}F^{-1}\left\{F_{f(t)}\left(x\right)^n\right\} = \frac{e^{-\frac{1}{2}x^2}}{\sqrt{2\pi}} \to \frac{1}{\sqrt{n}}F_{f(t)}\left(x\right)^n = e^{-2(\pi x)^2}$$

$$F\left(0\right) = 1\ ;\ F'\left(0\right) = 0\ ;\ F''\left(0\right) = -4\pi^2$$

$$\left(F_{\frac{1}{n^{\frac{4\pi^2}{2n\cdot 2}}}f(t)}\left(x\right)\right)^{2\frac{n}{4\pi^2}} = e^{-x^2}$$

$$\left(\frac{F_{f(t)}\left(x\right) - F'\left(0\right)\frac{x}{\sqrt{n}}}{F\left(0\right)}\right)^{2\frac{n}{4\pi^2}} \to e^{-x^2}$$

Following the desired form as above. Further, a nice generalization from Gaussians to Exponentials is the following;

$$\left(\frac{f\left(\frac{x}{a}\right)}{f\left(0\right)}\right)^{\frac{-a\,f(0)}{f'(0)}}$$

10 Lazy Gram-Schmidt

Suppose, again, that we wish to construct f from a set of basis functions ϕ_k. Regardless of if the ϕ's are orthogonal, we can take the following as a least squares reconstruction;

$$f\left(x\right) \sim \sum_k c_k\phi_k \to f\left(x\right)\cdot\phi_j \sim \sum_k c_k\left(\phi_k\cdot\phi_j\right) = M\vec{c} \to f\left(x\right) \sim \vec{\phi}\left(x\right)\cdot M^{-1}\left(f\left(x\right)\cdot\phi\right)$$

Where M represents the matrix of inner-products ($M_{ij} = \phi_i \cdot \phi_j$). It is amusing to observe that, for a general functional basis, the above triviality leads to a 'lazy Fourier Series', where a new orthogonal basis of $v_k(x)$ are constructed from the unit vectors $(\hat{\imath}, \hat{\jmath}, ...)$ of ϕ space;

$$v_k = P\phi_k\ \to v_1^T PDP^T v_2 = \phi_1^T D\phi_2 = 0$$

To give credence to the looser original formulation, f can now be expressed via an orthogonal decomposition, and then subsequently back into the original basis;

$$f(x) \sim \sum_k v_k(x) \frac{f \cdot v_k}{v_k \cdot v_k} = \sum_k \frac{1}{D_{kk}} \sum_j P_{kj} \phi_{k[j]}(x) \sum_i P_{ki}(f \cdot \phi_k)$$

$$= \sum_{j,k,i} \phi_{k[j]}(x) P^T_{jk} D^{-1}_{kk} P_{ki}(f \cdot \phi_k) = \phi(x)^T M^{-1}(f \cdot \phi)$$

In Gram-Schmidt, these $v_k(x)$ would be triangular in nature, rather than the 'lazy' method of a full linear combination of all ϕ_k.

As simple as least-squares reconstruction is, it is nonetheless fascinating to observe the behavior of the lazily-orthogonal v_k. The set of shifted Gaussians, for example, on $(0, \infty)$ is particularly striking; being transformed into a nearly sinusoidal basis.

So, the next time you seek to minimize the functional;

$$\int \left(f(x) - \sum_k c_k \phi_k(x) \right)^2 dx$$

Check out the orthogonal basis of $P\phi_k$ your basis creates.

11 Substituted Fourier Series

The standard Fourier expansion of;

$$f(t) = \sum_{n \in \mathbf{Z}} e^{-int} c_n \ ; \ c_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(z) e^{inz} dz$$

Has a trivial substitution with non-trivial effects, as for any 'nice' function L the following also exists;

$$f(t) = \sum_{n \in \mathbf{Z}} e^{-inL(t)} c_n \ ; \ c_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} f\left(L^{-1}(z)\right) e^{inz} dz$$

In a paper by *Elaissaoui and Guennoun* as well as one by *Patkowski* the function $L(t) = \frac{1}{2} \tan\left(\frac{t}{2}\right)$ is used with great success in producing a Fourier series which converges on the whole Real line as a tool in studying $\zeta(s)$. Here, we will be much less productive, and note that by taking

$$f(t) = \sum_{n \in \mathbf{Z}} t^{in} c_n \ ; \ c_n = \frac{1}{2\pi} \int_{e^{-\pi}}^{e^{\pi}} \frac{f(z)}{z} z^{-in} dz$$

We have now generated a 'Taylor series' in complex powers, with convergence occurring only in the interval $[e^{-\pi}, e^{\pi}]$. The above formula for c_n is of course equivalent to taking the Laurent expansion of $f(t^i)$.

12 Solving Elastica with Paper

Suppose you are driving, walking, shoveling snow, lawn-mowing, swimming, or any other activity where a turn must be made and where pivots, sharp accelerations, or bends are undesirable. Which curve minimizes the stress of this turn?

Suppose we wish to turn 90 degrees to the right over the course of one second over a lateral distance of one meter. We being at $x(0) = 0, y(0) = 0$ and take an upwards initial velocity $x'(0) = 0, y'(0) = 1$ and subsequently end at $x(1) = 1, y(1) = 0$ with $x'(1) = 1, y'(1) = 0$. Which parameterization $x(t), y(t)$ will minimize the amount of steering effort on our car?

The true solution to this family of variational minimization problems, known as the *Elastica Problem*, formally involves the minimization of $\int \kappa(s)^2 ds$, for κ the curvature, as this is proportional to the total energy held in a deformed elastic beam. While known solutions exist, the following approximate method is presently more entertaining;

Taking out a strip of paper and holding the endpoints at the initial positions and gently oriented with the correct velocities will naturally produce a minimizer of elastic potential energy subjected to the optimization constraints, in turn giving an optimizer of a curve with the the softest curvature, ultimately yielding the solution path.

Should you need to complete a 180 degree turn, simply reposition your grip about the paper to match the terminal velocities and an additional kink will be added to satisfy the conditions.

And they say paper road-maps aren't useful...

13 Remarks on a Matrix Exponential

In *Computing Integrals Involving the Matrix Exponential*, Van Loan identifies, among numerous other results, the relationship between;

$$\exp\left(\begin{pmatrix} -A^T & Q_0 \\ 0 & A \end{pmatrix} \Delta\right) \Leftrightarrow \int_0^\Delta e^{A^T t} Q_0 e^{At} dt$$

For constant, square matrices A, Q_0. The above procedure is important in Kalman Filtering, for example, where the effect of noise in the derivative space Q_0, along with the linear dynamics $\dot{x} = Ax$, can be be integrated into their resulting quantities in state space.

The beauty of Van Loan's methods demonstrate a connection between the matrix exponential, derivative matrices, and, of course, integration. Where else can these relationships lead?

In the analog of single-variable functions, for instance, we make the following *Statement:* Let;

$$A = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & a & 1 & 0 \\ 0 & 0 & a & 1 \\ 0 & 0 & 0 & a \end{pmatrix}$$

Then;

$$\int_0^\Delta e^{at} Q(t) dt \approx \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}^T e^{A\Delta} \begin{pmatrix} 0 \\ Q(0) \\ Q'(0) \\ Q''(0) \end{pmatrix}$$

For sufficiently small Δ, and with the pattern continuing for all counts of derivatives / matrix dimensions. For $a = 0$, all we need is the clear nilpotency of A, giving;

$$\int_0^\Delta Q\left(t\right) dt \sim \sum_{k=0}^{n-2} Q^{(k)}\left(0\right) \frac{\Delta^{k+1}}{(k+1)!} = \left\{1, \Delta, \frac{\Delta^2}{2}, \frac{\Delta^3}{6}...\right\} \cdot \left\{0, Q\left(0\right), Q'\left(0\right), Q''\left(0\right)...\right\}$$

$$\left[e^{A\Delta}\right]_{0,j} = \sum_{k=0}^{n-2} \left[A^k\right]_{0,j} \frac{\Delta^k}{(k)!} = \frac{\Delta^j}{j!} \to \left\{1, \Delta, \frac{\Delta^2}{2}, \frac{\Delta^3}{6}...\right\}$$

As aforementioned, it is enlightening to look at the system dynamics;

$$\frac{d}{dt}\begin{pmatrix} x \\ x' \\ x'' \\ ... \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & a & 1 & 0 \\ 0 & 0 & a & 1 \\ . & . & . & . \end{pmatrix} \begin{pmatrix} x \\ x' \\ x'' \\ ... \end{pmatrix} \Rightarrow x = e^{At}_{\text{Row } 0}$$

It is straightforward to demonstrate that this first row (all that is necessary given the first vector multiplication of mostly zeros) is composed of terms of the form;

$$[e^{At}]_{0,k} = x_k = e^{at} I^k \left\{ e^{-at} \right\} \Rightarrow x_k' = a x_k + x_{k-1}$$

Which follows precisely the chain of derivatives established in the system. Expanding out Q into a Taylor series about 0 and examining the integrals of the form $\int_0^\Delta x^k e^{ax} dx$ is sufficient to see how the general case arises.

I find it most delightful how the matrix exponential appears to furnish the right ingredients to complete the Taylor series of Q, concluding an *approximation* of a class of integrals. None of this is particularly surprising given that we have constructed the matrix exponential to be the solution to a trivially truncated differential equation describing our target function.